UTSAV CHAUDHARY

Texas | utsavmaan28@gmail.com | (607) 296-9583 | thatinsaneguy.com | in/utsavmaan28 | github/UttU28

Computer Science graduate supporting **warehouse automation** software at live **distribution centers**: **deployment** and **go-live** support, **SIT**/**UAT** testing, **Linux** log analysis, **SQL** data validation, and **Python** automation across **AMR** fleets and **WMS** integrations, with **JIRA**/**ServiceDesk** documentation and direct customer support.

Technical Skills

Warehouse Automation & Robotics: AMR, AGV, warehouse automation, WMS integration, GM Tower, SIT, UAT, test case execution, go-live support, hypercare, fulfillment and distribution sites, barcode management

Linux, Cloud & Debugging: Linux/Unix, SSH, log analysis, command-line debugging, systemd, root cause analysis, escalation, Azure, AWS, Docker, Kubernetes, firmware, bot error recovery, maintenance cage, bench debugging

Scripting, Data & APIs: Python, SQL, REST API testing, request/response validation, data validation, Git/GitHub, automated testing, test scripts, Python automation

Tickets & Coordination: JIRA, ServiceDesk, issue and resolution documentation, customer communication, remote support, Agile, nights, weekends, shift coverage, customer-site travel

Experience

Associate Engineer, Field Implementation, Hybrid, **MidAmerican Energy**, TX — Feb 2025 - Present

- Supported **software implementation** and **go-live** for a **300+** unit **AMR/AGV** fleet running **GM Tower**, assisting **System Integration Testing (SIT)** and **User Acceptance Testing (UAT)** on each release and staying through post-production **hypercare** across peak waves.
- Executed **test cases** and validated system functionality across the fleet and upstream **WMS** integrations, recovering **40+** live **bot errors** per shift with **Linux** and **SSH log analysis**, and escalated recurring faults with **root cause** data to cut repeat failures **30%**.
- Wrote **SQL** queries to validate task, inventory, and **barcode** records against fleet data and investigate mismatches, keeping **100%** serial-to-unit traceability across **12** months with zero record mismatches at audit.
- Ran and modified **Python** scripts for fault reporting, reconciliation, and automation, and logged **20+ JIRA**/**ServiceDesk** tickets per week to closure with issues, resolutions, and procedures documented for the next shift.

Implementation Support Engineer, On-Site, **Chevron**, GA — Jan 2024 - Jan 2025

- Supported on-site **software deployment** and customer **go-live** at field operations sites, executing **SIT** and **UAT** test cases and validating **REST API** request/response behavior ahead of **AKS** promotion, maintaining **99.9%** availability.
- Analyzed application and **telemetry** logs on **Linux** via **SSH** to troubleshoot integration issues, cutting mean time to resolution from **6 hrs** to **2 hrs** and documenting each **root cause** in **JIRA**.
- Modified and ran **Python** test and automation scripts, validated operational data with **SQL** queries against **Azure SQL**, and kept **Git** versioning on scripts, configs, and test data.

Projects

Self-Hosted Private Cloud — Personal Project

- Designed a private cloud on a custom server rack and replaced paid streaming and cloud storage with self-hosted **Jellyfin**, **Immich**, and **Nextcloud** for media, photos, and files on the home network.
- Wrote **Infrastructure-as-Script** pipelines (**deploy.sh** and per-service scripts) for **Docker Compose**, **PM2**, **nginx**, and **Certbot** so one command deploys the stack with **HTTPS**, and again at boot with **systemd**.
- Run the host on **Arch Linux** with an **Intel Core i7-14700HX**, **32 GB** RAM, an **NVIDIA RTX 4070**, and a **1 TB** NVMe, and keep service data on a **2 TB** disk mirrored daily at 4 AM with an **rsync systemd** timer.
- Built a **FastAPI** dashboard on **PM2** for **10+** projects, with process, **Docker**, port, and **HTTPS** checks plus one-click restart, and run a **kind** lab for Deployments, Services, Secrets, and Ingress.

Automated Ambu Pump (Open-Source COVID-19 Response)

- Built an **automated Ambu pump** with an open-source team during **COVID-19** for **hands-free ventilation**.
- Designed the mechanical drive with **3D-printed** gears and enclosure, **DC motor** and **servo** actuation, and tuned the squeeze cycle for a steady breaths-per-minute rate through hours of continuous running.
- Wrote the **Arduino** control code in **embedded C++** for adjustable rate and depth, and validated squeeze pressure and timing on the bench with the team before each unit was shipped.

Education

State University of New York at Binghamton — Aug 2022 to Dec 2023

Master of Science in Computer Science — 3.4/4.00 GPA